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ON THE LEAST SHARE IN THE EXCHANGE OF MANA

By SELBY AND ORMES

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ON THE LEAST SHARE IN THE EXCHANGE OF MANA

By Selby and Ormes*

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Summary. Classical nomological algebra treats the exchange of mana between the Web of Laws and the depths of Tiamat as a quantity divisible without end. From more than four centuries of comparison registers we recompute the "order residual", and find that within the precision of the registers it does not vary with instrument, environment or epoch. Writing the exchange as two operations, raising and lowering, and supposing the difference between their two orders of application to be a constant greater than nought, we prove under some general rules of the algebra that a definite amount of exchange can only be a whole multiple of a least share; we compute its value from the registers and, following the casters' old usage, call it a wisp. Three predictions are offered for testing by others with different instruments.

§ 1. Introduction

Severin's Eighth Question [1] reads: "Is mana continuous?" In F.E. 307 the Millennium Foundation set it as the fifth of the Millennium Seven [7], asking that a least, indivisible share in the exchange of mana be proved, its value computed, and the result confirmed by independent measurement.

Casters have always said that mana comes "wisp by wisp". We borrow the name, but do not take the intuition as evidence. Besides intuition, three approaches have been tried before, without result. The first was to ask casters skilled in fine control to divide mana again and again and to see where it would divide no further; but whether the limit lay in the mana or in the caster's hand, there was no telling. The second was to consult the station mana-meters, whose readings have been smooth for centuries since Fanshawe drew up the rules for stations about C.R. 950; but the finest division of these meters is far coarser than the share in question, and smoothness proves nothing of continuity. The third was to argue from classical nomological algebra; but there the amount exchanged is written as a continuous quantity from the start, so that continuity is not concluded but assumed.

Classical nomological algebra, since Elworthy [3], writes the exchange between the Web of Laws and the depths of Tiamat as a continuous quantity. Successive application in that algebra mostly depends on order; yet the two operations of exchange, raising and lowering, have always been treated as a pair that may be interchanged, raising then lowering giving the same as lowering then raising. Classical learning, treating mana as continuous, computed wards and tides in agreement with observation, and continuity came to be taken for granted.

Our point of departure is a very small number. When the Seventh Division compares its standard crystals, receiving before releasing gives a slightly different result from releasing before receiving; the difference is entered in the registers as the "order residual", and has always been taken for instrumental error. In Part I (§§ 2–5) we show that it is not instrumental error: within what the registers can distinguish, it does not vary with instrument, environment or epoch. In Part II (§§ 6–10) we deduce from it that the exchange is not

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continuous, compute the least share, and show why the classical theory nevertheless holds. In Part III (§§ 11–13) we propose tests, and discuss the bearing of the result upon the standards of measure and the questions it leaves open.

PART I. THE REGISTERS

§ 2. The procedure of comparison

Under the classical procedure, when a city's secondary standard is compared with the primary, two sets of standard crystals [2] are placed side by side on the comparing bench and joined through a single conducting crystal to the Web of Laws. The comparison proceeds by small shares received and released alternately: at one opening and closing of the conducting crystal each side, primary and secondary, receives a share of mana, and at the next each releases one; or the reverse, releasing first and receiving after. One receiving and one releasing make an alternation; after an alternation both sides return to their former charge. Alternations are of two orders, receiving before releasing and releasing before receiving; the procedure takes them by turns, half of each, so as to cancel the drift of the crystals during the comparison, and one alternation of each order makes a pair. A comparison runs to some thousands of alternations. The bench has two counting dials: the difference between the two sides in the alternations receiving first runs on to one, and in those releasing first on to the other, neither being cleared during the comparison; at its end each is read once and entered in the register as a matter of routine. The procedure regards the difference between the two orders as due to drift and dissipation, averages it away at the end, and takes only the averaged result as the calibration of the secondary.

The order residual has thus stood in the registers for centuries without anyone treating it as a quantity. In a single comparison it amounts to no more than a fraction of the smallest division, and to those making the comparison it was not to be distinguished from an error of reading.

The registers of the Seventh Division were for the most part burned in the Great Tide, and those that survive begin about C.R. 1450; the registers of secondary standards kept by seven cities begin and end at various dates. We have recomputed the whole of these four centuries of records. In recomputing we have used only the original readings, never the results calculated at the time; from each comparison the supply, the exchange with the surroundings and the dissipation have been deducted, the dissipation being calculated by the general account established since Finch [2]. The records of the First Era have all been reduced, by the numbers and calibration dates of the standards entered in the registers, to the master standard fixed by the Seventh Division. We found in all some 21,900 comparisons; the 2,100 and more that cannot be so reduced, about one in ten, are not used for the value, but only to see the form of the residual, which is the same as in the rest. They lie mostly in the first century of the First Era and in the cities of Tallin and Onais. The comparisons admitted number 19,762: 612 of the Seventh Division before the Great Tide, used for the comparison across the Tide in § 5, and 19,150 of the seven cities in the First Era, used for the value of the wisp.

The recomputation followed the old rule of great reckonings, that two places reckon apart. The University of Kanaich and Tumen University each, with their own hands and without communication, copied from the original registers, reduced and combined; only

results that agreed group by group between the two were admitted. The three hundred and more comparisons on which they disagreed were copied afresh from the registers, and those that still disagreed were not used; they are counted within the one in ten above.

§ 3. The order residual

The order residual of a comparison is the sum of what is recorded for all its alternations receiving before releasing, less the sum for all those releasing before receiving, and is reckoned positive when the former is the greater; the number of reversals is the number of pairs into which the two orders are joined in the comparison. The recomputation shows three things.

(i) The residual always lies to one side, and is not positive and negative at random: of more than nineteen thousand comparisons fewer than twenty give a negative residual, all of them comparisons of very few pairs, whose residual is no greater than the error of estimating the reading.

(ii) The residual is proportional to the number of reversals of order in a comparison: each reversal adds the same amount (Fig. 1). The duration of a comparison, the number of its alternations and the total passed through the conducting crystal all increase with the number of pairs; but the dissipation and drift they bring are the same for both orders, and cancel in the residual. The two procedures of the First Era differ threefold in the duration of an alternation, yet the amount added by one reversal is the same in both (Table 2, § 4), so that the residual does not depend on duration.

(iii) The amount added by one reversal does not vary with the city, with tide years or ordinary years, or with the manner of coupling (Table 1); grouped by the three materials of the standard crystals, the three groups lie between 3.46 and 3.48, each with an error of about 0.02, and show no difference either.

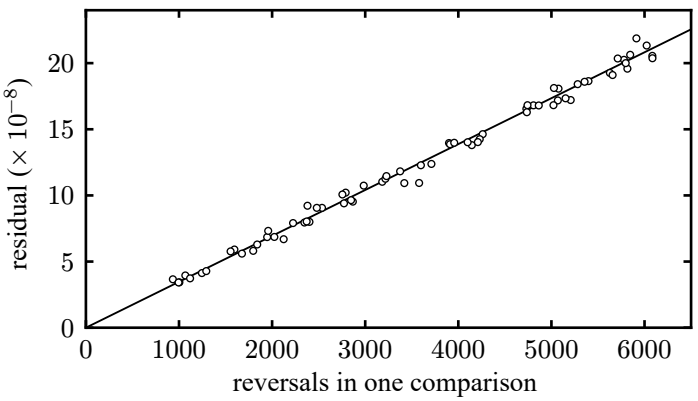


Fig. 1. The order residual of each comparison against the number of reversals. Each point is one comparison from the First-Era registers of the seven cities; the line is fitted without being required to pass through the origin, its intercept differing from nought within the error, and its slope is the amount added by one reversal.

TABLE 1. Amount added by one reversal (records after the Great Tide only, in the old unit of exchange fixed by the master standard).

Group	*Comparisons*	*Amount per reversal (× 10⁻¹¹)*
Kanaich	4,120	3.47 ± 0.02
Stormthrone	2,980	3.46 ± 0.03
Lataki	3,310	3.48 ± 0.03
White Tower	2,450	3.47 ± 0.03
Varen	2,760	3.46 ± 0.03
Tallin	1,890	3.49 ± 0.04
Onais	1,640	3.47 ± 0.04
Tide years	3,920	3.47 ± 0.03
Ordinary years	15,230	3.47 ± 0.01
Direct coupling	9,870	3.47 ± 0.02
Coupling through a mana engine	9,280	3.47 ± 0.02

Dissipation and instrumental error both vary with these conditions; the residual does not. Every ± in the tables is the error of the group's mean, which includes, besides the scatter of readings, the errors of reducing that group's secondary standards and of deducting its dissipation, and is therefore larger than the scatter alone; the first entry of Table 4 reckons the scatter of readings only. Tide years furnish a fifth of the comparisons because standards are easily disturbed by the tide, and the procedure requires the cities to compare more often about a tide year. The secondary standards of all the cities are reduced to one master standard, whose error is common to every group and cannot be seen by grouping; it is entered in Table 4. Tables 1 and 2 are different divisions of one body of records, used only to see whether the residual varies with conditions; the value of the wisp is computed once from all the records, and the groups are not combined again by weight.

§ 4. Tests of exclusion

Several further possibilities must be excluded one by one (Table 2).

First, an asymmetry in the procedure itself. Were the two orders carried out differently in duration or in manner, the residual might arise from the procedure. The procedure of comparison has been written in four forms in the classical age and the First Era, differing in the duration of an alternation, the number of pairs in a comparison and the manner of alternation; recomputed form by form, the two procedures of the First Era give the same amount for one reversal, and the two classical ones a slightly higher, for which see the note below the table.

Secondly, the habits of hands. Comparisons are mostly conducted by officers sent by the Seventh Division, and a habit common to them might produce the residual. Recomputed by the officer in charge, the three hundred and more officers show no difference among themselves; Table 2 gives only three groups by length of service.

Thirdly, the charge of the crystals at the time. If the residual depended on how full the crystals were, it would not be a property of the exchange itself. The charge at a comparison ranges from nearly full to nearly empty; recomputed in four classes by the charge at the start of the comparison, the amount added by one reversal is again the same.

Fourthly, an asymmetry between the two sides of the bench. In about a quarter of the comparisons the primary stood on the right of the bench, and in the rest on the left; recomputed by the two placings, the amount added by one reversal is the same.

Table 2. Recomputation by procedure, officer and charge (amount per reversal, × 10^{-11}).

Group	*Comparisons*	*Amount per reversal*
Classical procedure A	412	3.50 ± 0.04
Classical procedure B	200	3.53 ± 0.06
First-Era procedure A	11,340	3.47 ± 0.01
First-Era procedure B	7,810	3.47 ± 0.02
Officers with under ten years' service	5,020	3.47 ± 0.02
Officers with ten to thirty years' service	9,660	3.47 ± 0.02
Officers with over thirty years' service	4,470	3.47 ± 0.02
Charge above three quarters	4,880	3.47 ± 0.02
Charge from one half to three quarters	5,130	3.47 ± 0.02
Charge from one quarter to one half	4,920	3.48 ± 0.02
Charge below one quarter	4,220	3.46 ± 0.03

The two classical procedures give slightly higher values because they were used only before the Great Tide and are reckoned in the old units of that time; see the next section. Procedure and epoch cannot therefore be separated. The classes by service and by charge contain the records of the First Era only, 19,150 in all.

So far as these groupings can distinguish, then, the residual shows no variation with instrumental error, dissipation, procedure, hands, charge or placing; a bias common to all the groups, such as the one master standard or the one kind of bench, cannot be seen by grouping. We interpret the residual as a property of the exchange itself; this interpretation still awaits the test of other colleges with other instruments (§ 11). It is consistent also with the records of the plane-gate of the White Tower, kept without interruption since about C.R. 1260: within the reach and precision of the gate, the course of the laws has scarcely varied in centuries; but the gate sees only the working of the web, and cannot by itself show that the exchange in the depths is unchanged.

§ 5. Before and after the Great Tide

Before and after the Great Tide the case is a little different (Fig. 2, Table 3). Read in the old units, the residual after the Great Tide is about one part in a hundred smaller than before. If the residual is a constant of the exchange, this change can only come from the unit itself; this is a conditional, and cannot be turned round to prove the residual a constant. Taunton showed about F.E. 20 [5] that the standards disturbed by the tide had suffered a lasting change in capacity and response: after the Great Tide the old unit fixed by the master standard was larger than before, the ratio of the unit after the tide to the unit before being 1.011 ± 0.002. He measured it against reference crystals kept where the tide had touched them least, independently of the registers used here. By this ratio the 3.51 before the tide becomes 3.47 in the unit after it, which agrees with the 3.47 after; and the ratio of Table 3, 1.012 ± 0.009, agrees with Taunton's within the errors. The simplest explanation is that it was the standard that changed, not the residual. Taunton urged in his day that the unit

of mana be defined by a quantity that does not change with the instrument, but no such quantity was then known; the order residual is such a quantity.

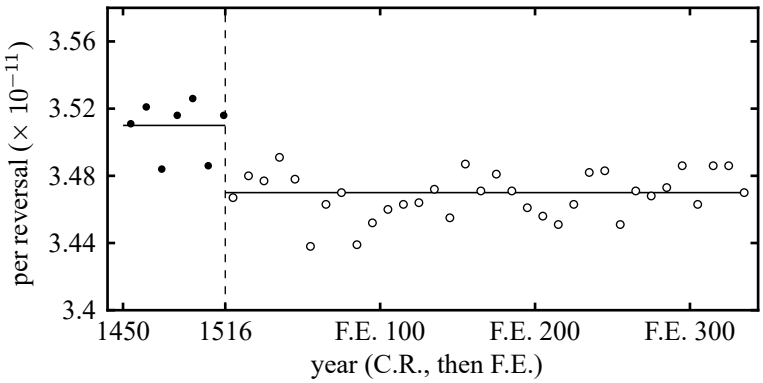


FIG. 2. The amount added by one reversal, decade by decade, each in the old unit of its time. Each point is the mean of ten years; the upright line marks the Great Tide (C.R. 1516), and the two horizontal lines are the means of the two stretches.

TABLE 3. Before and after the Great Tide.

Record	*Amount per reversal (each in the old unit of its time, × 10⁻¹¹)*
C.R. 1450–1515, registers of the Seventh Division, 612 comparisons	3.51 ± 0.03
F.E. 1–337, the seven cities, 19,150 comparisons	3.47 ± 0.01
Ratio, before to after	1.012 ± 0.009
Lasting change of the master standard's unit measured by Taunton (unit after the tide to unit before, about F.E. 20)	1.011 ± 0.002

PART II. THEORY

§ 6. Exchange in nomological algebra

Nomological algebra calls the condition of the web at a place a web-state, and operations act upon web-states. We write XY for the operation of applying Y first and X after. In nomological algebra joint application answers to addition and successive application to multiplication; successive application is associative and distributes over joint application; a number and an operation may be multiplied in either order; and the identity operation changes nothing. These are general rules of the algebra since Elworthy [3], holding among exchanges that are compatible with one another, and the raising, lowering and amount of this paper lie within that range. They are all we shall need, together with the rules of the null at the end of this section.

Nomological algebra writes the exchange by two operations: raising, U, which raises one share of mana from the depths of Tiamat into the Web of Laws; and lowering, D, which lowers one share from the web back into the depths. When a crystal receives or releases mana in comparison, the exchange always passes between the Web of Laws and the depths

of Tiamat: in receiving, mana from the depths is raised into the web; in releasing, mana in the web is lowered into the depths. Receiving before releasing is therefore raising before lowering, written DU; releasing before receiving is lowering before raising, written UD.

We take further a general rule of nomological algebra since Elworthy [3]: U and D are dual to each other, and an operation and its dual applied in succession, in either order, yield an amount that is never negative.

The mana a web-state has drawn from the depths and not yet returned we call the amount of exchange of that web-state, or simply its amount, Q. In the usual way of writing amounts of exchange in nomological algebra, it is lowering followed by raising:

$$Q = UD \tag{1}$$

Two things must also be kept apart. A web-state whose amount is nought is still a web-state; an operation which yields nothing at all we say yields the null. The null is not a web-state, and is not the same as a web-state of amount nought. Any operation applied to the null yields the null; the null applied jointly with any web-state yields that web-state; and a number multiplied into a web-state yields the null only if the number is nought. Raising or lowering a web-state yields, if not the null, again a web-state; and a joint application acts upon a web-state term by term.

That a web-state has the definite amount q means that Q applied to it yields q times the web-state itself. How the readings of the registers answer to these operations we take as follows. One opening and closing of the conducting crystal is, in nomological algebra, one application: receiving is one application of U, releasing one of D. The difference between the two sides recorded on the bench for an alternation, apart from the difference between secondary and primary themselves, is by the usage of reading since Elworthy [3] the excess of the pair of operations over the amount: if the pair, applied to a web-state of definite amount q, multiplies it by a number exceeding q by so much, so much is recorded. Both sides being reset after each alternation, the two alternations of a pair are made upon the same web-state; the difference between secondary and primary does not depend on the order, and cancels in the difference of the two. The difference recorded within a pair, receiving before releasing less releasing before receiving, is therefore the difference between the numbers by which DU and UD multiply that web-state; and the residual is the sum of these differences over the pairs. That one opening and closing is one application, and this manner of reading, are hypotheses of this paper; the tests of § 11 are tests of them also. On this reading one opening and closing exchanges a single wisp; the "share" of the classical procedure was always far below a division, and no one knew how small it was.

§ 7. The relation of successive application

The result of § 3, written as a formula, is

$$DU - UD = \varepsilon \tag{2}$$

where ε is the amount added by one reversal, and the right-hand side is the product of ε with the identity operation, written simply ε. What the registers record are readings made under a particular procedure; to extend them to a relation (2) holding for every web-state is an hypothesis of this paper. It is supported by two findings of §§ 3–4: the residual shows no variation with the charge of the crystals at the time, nor with the procedure, the city or the

epoch. By § 3 (i) the residual lies always to one side, receiving before releasing recording more than releasing before receiving, so that ε is greater than nought. The classical theory, in assuming that the two operations may be interchanged, assumed in effect that ε is nought.

§ 8. The theorem of shares

From (2) it follows that

$$QU = U(Q + \varepsilon), \quad QD = D(Q - \varepsilon) \tag{3}$$

For $QU = UDU = U(DU) = U(UD + \varepsilon) = U(Q + \varepsilon)$; and $D(Q - \varepsilon) = DUD - \varepsilon D = (UD + \varepsilon)D - \varepsilon D = UDD = QD$.

The meaning of (3) is this: if a web-state has amount q, the web-state obtained by applying U once has amount $q + \varepsilon$, and that obtained by applying D once has amount $q - \varepsilon$, unless it is the null.

THEOREM 1. *If the amount of a web-state is a definite number, it can only be one of* $0, \varepsilon, 2\varepsilon, 3\varepsilon, \ldots$*; and if there is any web-state of definite amount, all of these values occur.*

Proof. Take a web-state of definite amount q, q being finite, and apply D to it repeatedly. By (3), so long as a web-state results, its amounts are $q - \varepsilon$, $q - 2\varepsilon$, …. Since ε is greater than nought, and by the rule of duality the amount is never negative, D cannot be applied without end: at some step the result is no longer a web-state but the null. Let the web-state before that step have amount q_0. For this web-state $Q = UD$; lowering gives the null, and raising the null gives the null again; so $q_0 = 0$. Hence q, less some number of ε, is exactly nought; that is, q is a whole multiple of ε. Conversely, apply U once to the web-state of amount nought just reached. The result is not the null: for by (2), raising it and then lowering is $(UD + \varepsilon)$ applied to it, which gives ε times the web-state itself, and is not the null; whereas had raising given the null, lowering would give the null again. By (3) its amount is ε. In the same way, raising and then lowering a web-state of amount q gives $q + \varepsilon$ times the web-state, which is not the null, so that U may always be applied again, giving $2\varepsilon, 3\varepsilon, \ldots$ in turn.

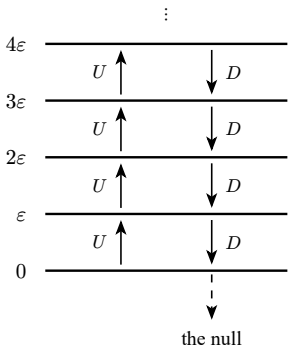


FIG. 3. The ladder of the theorem of shares (a diagram). Each rung is a definite amount; U raises by one rung and D lowers by one; lowering the web-state of amount nought gives the null.

The ladder of the algebra has no top; how far an actual instrument can climb it is another matter. The theorem concerns only web-states whose amount is a definite number. We

assume further that before and after a single exchange the web-state joined to the conducting crystal is of this kind, so that the amount of a single exchange is a whole number of wisps. The mean of many exchanges need not be: if half are nought and half one wisp, the mean is half a wisp. This assumption is put to the test by the counting of jumps in a weak field (§ 11).

Under these hypotheses, therefore, the exchange of mana between the Web of Laws and the depths of Tiamat is not continuous. It proceeds share by share, each share being ε, and there is no half-share: ε is the least share in the exchange of mana. Casters have always said that mana comes wisp by wisp, and following their usage we call ε a wisp.

Severin's Eighth Question can thus be answered: as regards the exchange between the Web of Laws and the depths of Tiamat, mana is not continuous. Of other quantities we say nothing.

§ 9. The value of the wisp

By §§ 3 and 7, a wisp is the amount added by one reversal. The calculation is this. The residual of each comparison equals the number of reversals multiplied by ε, together with the bias left by dissipation and instrumental error; all the records are combined once by the doctrine of errors into a weighted mean, the weights being set by the number of pairs and the error of estimating the reading in each comparison, and the errors of reduction from the old units being included. The relation (2) is exact; the uncertainty of the value arises from the old units and the records of past ages (Table 4). Combining all the comparisons of the seven cities in the First Era, we find

$$\text{one wisp} = (3.47 \pm 0.01) \times 10^{-11}$$

old units of exchange (after the Great Tide, as fixed by the master standard).

Table 4. Sources of error in the value of the wisp ($\times 10^{-11}$).

Source	*Error*
Scatter of the readings of single comparisons	0.005
Reduction of the cities' secondary standards to the master standard	0.006
Deduction of dissipation	0.004
Differences between procedures	0.003
Total	0.009

The total is the square root of the sum of the squares, and is here taken as 0.01. So to combine them is to treat the four sources as independent; the correlation arising from the one master standard shared by the cities has been put into the second.

The instruments of the comparing procedure have a smallest division of about ten thousand wisps, and the finest weak-field sensing piece now made resolves no better than some three wisps at a reading. This is why a wisp has never been read directly.

How a wisp is to be got from divisions of ten thousand wisps may be shown by a single comparison. A comparison has commonly two or three thousand pairs; with two thousand five hundred, the residual is some two thousand five hundred wisps, a quarter of a division; and those comparing estimate the reading, by custom, to a twentieth of a division, some five hundred wisps. A wisp found from this one comparison has an error of about one fifth. Combined over nineteen thousand comparisons, the error falls to about one part in seven

hundred, or 0.005×10^{-11}, the first entry of Table 4. What no single comparison can show, four centuries of registers show together.

§ 10. The macroscopic limit

An exchange on the large scale amounts to thousands of millions of wisps; a difference of one wisp lies far below the error of measurement, UD and DU are scarcely to be told apart, and to treat the exchange as a continuous quantity does no harm. If the amount of a web-state is one old unit, the difference of one reversal is about one part in thirty thousand millions of it. In general, if the amount at a place is q, the ratio of the difference of order to the amount is ε/q; when q is many wisps this ratio tends to nought, and (2) returns to the classical form, in which the two operations may be interchanged.

Classical nomological algebra and the tide equations are therefore not overthrown by this paper. The difference of one reversal bears to the amount the ratio ε/q, which may be neglected when the amount is many wisps. How the tide equations as a whole should be amended would require a passage from the exchange wisp by wisp to the continuous equations, which this paper has not made; we can say only that observation on the large scale still agrees with the continuous theory. Only where mana is extremely weak, or gathers and disperses violently within very small regions, may the continuous theory fail; where the limits of its failure lie, we do not venture to predict.

PART III. PREDICTIONS AND REMARKS

§ 11. Predictions that may be tested

Our deduction used the registers and a general rule of nomological algebra. Whether the problem of the least mana has been solved must be settled by other colleges with other instruments. The three methods of measurement below are all independent of the registers used here.

(i) Counting jumps in a weak field. In a still room far from any place of casting, out of reach of the interference of casting nearby, let the mana from without be reduced to extreme weakness; no room keeps out the tide, and the rise and fall of the tide are to be recorded at the same time by a second sensing piece not joined to the web, and deducted. Let a sensing piece resolving better than half a wisp be joined to the Web of Laws and the exchange be read continuously. After calibration, and when the response of the sensing piece has been allowed for, the readings should jump step by step, each step being one wisp or a whole multiple of one (Fig. 4). With other crystal materials, other manners of coupling and other supplies the height of the steps should not change; the finer the sensing piece, the clearer the steps, but their height should still not change.

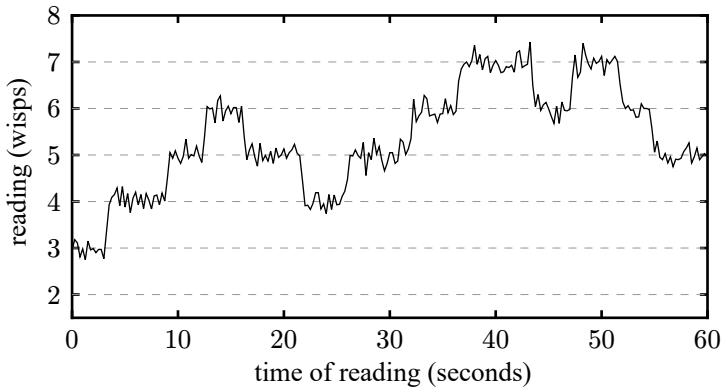


Fig. 4. Predicted readings in a weak field (a diagram, not an observation). With a sensing piece resolving half a wisp the readings should jump between whole multiples of a wisp; the broken lines mark those multiples.

(ii) Resonance in a cavity. Students of arrays have long calibrated the resonant frequency of an array cavity by the mana stored in it: with large amounts the shift of frequency is proportional to the amount stored, the proportion depending on the dimensions and material of the cavity. Whether the proportion still holds at the scale of a single wisp we cannot warrant; this method tests the law of the cavity as well, and the proportion should first be calibrated with amounts lowered step by step, to see whether it changes. If it does not, then when mana is let into the cavity very slowly the frequency should shift step by step, each step answering to one wisp or a whole multiple of one, the interval being this proportion multiplied by a wisp.

(iii) Slow discharge of a mana core. A mana core discharged very slowly through a very fine mana engine should yield its energy in steps, each answering to a wisp or a whole multiple of one. The procedure of Northcote [4] gives only the energy a core delivers when drained; between core capacity and total exchange there is as yet no common reckoning. For this method the relation between the two must first be calibrated afresh for one mana engine under conditions set down, the losses within the engine being deducted, before the core capacity answering to one step can be found.

There is also a negative prediction: after calibration, and when the response of the instrument has been allowed for, the net amount of every single exchange should be a whole multiple of a wisp, and there should be no half-wisp. The raw readings of an instrument may show changes smaller than a wisp, by reason of the duration of a reading, the response of the sensing piece or a divided path; these are no counter-instance. Only an event that is certainly an exchange of the kind measured, and is not a whole multiple of a wisp, is a counter-instance. If any method, after calibration and the exclusion of missed and merged steps, should find steps whose height varies with the instrument, or does not agree with a wisp, or should find exchanges certainly smaller than a wisp, the doctrine of this paper cannot stand.

§ 12. The wisp and the standards

Taunton urged about F.E. 20 that the unit of mana should be defined by a quantity that does not change with the instrument. Should this paper be confirmed by independent

measurement, the unit of total exchange might be defined as a fixed number of wisps, and the concentration of free mana and the capacity of mana cores might both be traced back to it, so that the three standards would at last rest on a common ground. The old records need not be discarded: provided the standard used and the date of its calibration are entered, they can be reduced in the manner of § 2. Whether this should be done we submit to the Seventh Division for preparation, and to the Academic Council of the Silver Society for decision.

§ 13. Remarks

First. This paper treats only of the exchange between the Web of Laws and the depths of Tiamat. Whether time, length or any other quantity is continuous we do not consider. Some may wish to explain the boundary of the tenth ring by a counting of wisps; for that we have no evidence.

Secondly. Why the difference of successive application is not nought, we cannot say. Westphal's doctrine of layers [6] gives the order of the layers, and the doctrine of seven layers supposes that the depths of Tiamat contain layers that strain, sort and divide; the wisp may arise in one of these, but that is conjecture, and this paper has no means of testing it.

Thirdly. The data of this paper are all taken from the registers of those who went before. Without readings entered as a matter of routine for centuries and never thrown away, we could not have begun; those who made the comparisons took this small difference for instrumental error, yet they entered it, item by item.

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